why is the transformation matrix not a tensor?

The transformation matrix $\Lambda^\mu{}_\nu$ does not transform like a tensor!

If you have a tensor, say $T^\alpha{}_\beta$, it transforms via

$$T'^\mu{}_\nu = \frac{\partial x'^\mu}{\partial x^\alpha} \frac{\partial x^\beta}{\partial x'^\nu} T^\alpha{}_\beta$$

↳ A Tensor's transformation rule depends on the Jacobian of the coordinate transformation.

$J^\mu{}_\nu = \frac{\partial x'^\mu}{\partial x^\nu}$: a matrix of partial derivatives

measures how the coordinate grid is stretched, squished, or sheared.

$\det(J) = 2 \rightarrow$ volume double

$\det(J) = 0 \Rightarrow$ dimension collapse.

$\det(J) < 0 \rightarrow$ orientation flips

The lorentz matrix $\Lambda^\mu{}_\nu$ is the jacobian of the coordinate transformation

$$x^\mu = \Lambda^\mu{}_\nu x^\nu$$

$$\Lambda^\mu{}_\nu = \frac{\partial x^\mu}{\partial x^\nu}$$

↳ $\Lambda^\mu{}_\nu$ transforms via group composition

$$\Lambda_{new} = \Lambda_2 \Lambda_1$$

↳ Tensors obey

$$T' = J T J^{-1}$$

There are 4 disconnected pieces in the Lorentz Group

① ② det → +1, → −1

③ ④ time direction → time-preserving, → time-reversing

$R \to$ rotations in 3D space $\quad R^T R = 1, \ (\det R)^2 = 1$

$\Lambda \to$ Lorentz in 3+1 rotation

$\det R = +1$ $\quad$ $\det R = -1$

"proper rotation" → describes ordinary motion

"improper rotations" reflection, → detects symmetry, parity.

There are 3 parameters you need to characterize 3D Rotations.

$$R(\alpha, \beta, \gamma) = R_{\hat{z}}(\gamma) R_{\hat{y}}(\beta) R_{\hat{x}}(\alpha)$$

$$= R_{\hat{n}}(\theta)$$

→ rotation about a certain axis $\hat{n}$.

$$\Lambda^T \eta \Lambda = \eta$$

→ take det of both sides

$$(\det(\Lambda))^2 = 1$$

$\Lambda = +1 \to$ proper Lorentz

$-1 \to$ improper Lorentz

The General Transformation matrix.

$$\Lambda = \begin{array}{c} \\ \\ \end{array}\begin{array}{c} \overbrace{\qquad\qquad}^{3} \\ \left(\begin{array}{c|c} & \text{boosts} \\ \hline \text{boosts} & \text{rotations} \\ & \\ & \end{array}\right) \end{array} \Big\}\, 3$$

6 parameters $\rightarrow$ 3 rot

$\searrow$ 3 boosts

$$\Lambda = R_{\hat{n}}(\theta)\, \Lambda_{\hat{b}}(v)$$

$\downarrow$ rotation around $\hat{n}$.

$\downarrow$ boost along $\hat{b}$.

Small transformations: Lie Algebra.

$$f(x+a) = \left(1 + a\frac{d}{dx} + \frac{a^2}{2!}\frac{d^2}{dx^2} + \dots\right) f(x)$$

$$f(x+a) = \underbrace{\cdots\cdots\left(1+\delta\frac{d}{dx}\right)\left(1+\delta\frac{d}{dx}\right)\left(1+\delta\frac{d}{dx}\right)}_{n\text{-steps.}} f(x) \approx f(x+\delta)$$

$$\delta = \frac{a}{n}$$

$$\lim_{n\to\infty}\left(1+\frac{a}{n}\frac{d}{dx}\right)^n f(x) = e^{a\frac{d}{dx}} f(x)$$

$$R_x(\alpha) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & \sin\alpha \\ 0 & -\sin\alpha & \cos\alpha \end{pmatrix} \xrightarrow{\text{for small } \alpha} 1 + \alpha \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}}_{\text{a generator of an infinitesimal rotation.}} + O(\alpha^2)$$

$$\begin{cases} \hat{L_x} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \\ \hat{L_z} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \end{cases}$$

$$\hat{L_y} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Can I come back to the og point?

$$T_1 = R_{\hat{y}}(\alpha) \quad , \quad T_1^{-1} = R_{\hat{y}}(-\alpha)$$

$$T_2 = R_{\hat{x}}(\beta), \quad T_2^{-1} = R_{\hat{x}}(-\beta)$$

$$R_y(\alpha) = e^{\alpha L_y} \quad , \quad R_x(\beta) = e^{\beta L_x}$$

$$R_x(-\beta)\, R_y(-\alpha)\, R_x(\beta)\, R_y(\alpha)$$

$$= (1-\beta L_x)(1-\alpha L_y)(1+\beta L_x + \dots)(1+\alpha L_y + \dots)$$

$$= 1 + \cancel{(\alpha-\alpha)} L_y + \cancel{(\beta-\beta)} L_x +$$

$$\underbrace{\alpha\beta \cancel{L_x L_y} + \alpha\beta \cancel{L_x L_y} - \beta\alpha L_x L_y - \alpha\beta L_y L_x}$$

$$\alpha\beta \underbrace{[L_x, L_y]} \rightarrow L_x L_y \neq L_y L_x$$

$$[L_x, L_y] = -L_z$$

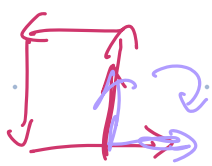
Order does matter

Infinitesimal boost:

$$\begin{pmatrix} \gamma & \gamma v \\ \gamma v & \gamma \end{pmatrix} \to 1 + v \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & & & \\ 0 & & 0 & \\ 0 & & & \end{pmatrix}$$

in the x-dir

2 boosts → an infinitesimal rot.



your orientation in space changes.

$\Lambda^0{}_0 > 0$ "orthochronous"

$\Lambda^0{}_0 < 0$ "antichronous"

Lorentz	proper	improper
ortho	$\mathbb{1}$	$P = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}$
anti	$\begin{pmatrix} -1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}$	$\begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix} = T$

	parity (→ reversal of space)	time-reversal
$t, \frac{\partial}{\partial t}$	1	-1
$\vec{r}, \vec{\nabla}$	-1	1
$\vec{v}$	$-\vec{v}$	$-\vec{v}$
ρ	ρ	ρ
$\vec{J}$	$-\vec{J}$	$-\vec{J}$
$\vec{F}$	$-\vec{F}$	$\vec{F}$
$\vec{E}$	$-\vec{E}$	$\vec{E}$
$\vec{B}$	$+\vec{B}$	$-\vec{B}$
$\vec{S}, \vec{L}$	$+\vec{S}, \vec{L}$	$-\vec{S}, \vec{L}$

pseudo vector

↳ figure out why it transforms that way!

$\nabla \times \vec{B} = \frac{\partial \vec{E}}{\partial t} + \vec{J}$

$\sim \underline{r \times p}$

using parity conservation to communicate w/ aliens

$\langle \vec{S}, \vec{p} \rangle$ → asymmetry of # of e^- along / oppose the spin.

	P	T
$\vec{\nabla}\cdot\vec{B}=0$	✓	✓
$\vec{\nabla}\cdot\vec{E}=\rho$	✓	✓
$\vec{\nabla}\times\vec{E}=-\frac{\partial\vec{B}}{\partial t}$	✓	✓
$\vec{\nabla}\times\vec{B}=\frac{\partial\vec{E}}{\partial t}+\vec{J}$	✓	✓

strong, electromagnetic, gravity

weak interaction ✗ time & parity